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September 23, 2026

Apple Machine Learning Research
Paris, France

Re: Internship – Machine Learning Research (200683768)

Dear Hiring Team,

I am applying for the Machine Learning Research internship at Apple MLR in Paris. I am a final-year Ph.D. candidate in computer vision at Linköping University, funded by ELLIIT and affiliated with WASP. My research is motivated by the observation that perception and planning tasks are often formulated as regression problems. I investigate how to accurately estimate and represent predictive uncertainty to build trustworthy learning systems and improve downstream decisions, particularly in safety-critical settings. This focus aligns closely with your work in uncertainty quantification, generative modeling, and foundational machine learning.

I began by developing Hinge-Wasserstein to represent multimodal aleatoric uncertainty without imposing a restrictive parametric distribution (CVPR Workshop, 2024). Through this work, I found that correctness prediction and uncertainty calibration require distinct metrics and favor different models. This finding prompted a systematic study of uncertainty metrics for deep regression (*Pattern Recognition Letters*, 2024). I then moved beyond uncertainty as an isolated prediction task to study its impact on downstream decisions. In RiskMonitor (2025), I adapted loss prediction to estimate collision risk from end-to-end driving models and support safer planning; this direction was directly inspired by Michael Kirchhof's work on loss-prediction-based uncertainty quantification. Most recently, MEOM (2026) uses complete 2D heatmaps rather than point estimates for multi-view human pose triangulation and assesses their reliability through HDR calibration. Together, these projects examine not only how uncertainty is estimated, but also how it improves the tasks that depend on it.

My current work extends this trajectory to calibrating multimodal outputs from vision-language-action models for automated vehicles, supported by a Vinnova FFI project and an upcoming WASP-funded research stay at TU Delft. This setting brings together the themes that have shaped my Ph.D.—multimodal prediction, calibration, and safety-critical perception—while opening a path toward more general learning systems.

I bring experience across the full research cycle: framing problems, developing methods in PyTorch, designing evaluation protocols, running multi-GPU experiments, and communicating results through publications. Through oral and poster presentations at international conferences, workshops, and summer schools, I have learned to communicate technical ideas clearly to diverse, cross-cultural audiences. Serving as primary advisor to nine master's students and as a reviewer for IJCV, CVPR, and AAAI has also strengthened my ability to give precise feedback and collaborate across projects. I would value the opportunity to bring this combination of methodological depth and application-driven thinking to Apple MLR, and to contribute research that is both scientifically rigorous and useful at scale.

Thank you for considering my application. I would be glad to discuss how my research in uncertainty-aware machine learning could contribute to your team.

Sincerely,
Ziliang Xiong